

DATA ANALYSIS ON DOMESTIC AIRLINES IN INDIA

SET ID - 222702



Presented By:

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ABSTRACT

- This study incorporates advanced statistical and machine learning approaches to build a prediction model for passenger travel in the changing environment of India's domestic aviation sector. On broad datasets, ARIMA and SARIMA analysis forecast passenger demand, payload factor, and resource allocation. Simultaneously, a Random Forest model predicts happiness based on in-flight amenities, reservation experiences, timeliness, and service quality using a heterogeneous dataset. The combination of ARIMA and Random Forest outputs provides a holistic methodology, revealing how expected passenger numbers correspond to consumer satisfaction levels. This report intends to give practical insights for improving India's domestic aviation business, supporting sustainable growth, and providing better customer experiences.



OBJECTIVE

- To predict the number of passengers travel in the domestic airlines using SARIMA Model
 - To forecast the customers satisfaction in domestic aviation by using Random forest
 - To finding the relationship between Predicted numbers of passengers and customer satisfaction to increase the passengers travel's in the domestic aviation in India.
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INTRODUCTION

- This paper examines the complex issues in the fabric of Indian internal aviation. Sophisticated analyses are critical for any sector that can be termed as an important factor in the economy of a nation or at any given level. This study aims at giving the contribution through development of a predictive model using machine as well as advanced statistical techniques. The intricacies of predicting passengers involve analysing historical data on traffic, changes within payload, and resource distribution. It further looks at the complex ties between estimated volume of passengers traveling and customers' joy; including comfort on flights such as refreshments provided, and promptness. Overall, this research aims at giving priceless knowledge towards the sustainability development of Indian domestic air travel sector maintaining both efficient operations & service oriented approaches.



LITERATURE OF REVIEW

Author

Khodijah Hulliyah

Review Paper

- Predicting Airline Passenger Satisfaction with Classification Algorithms
- [IJIS International Journal of Informatics and Information Systems](#) 4(1):82-94



RESEARCH METHODOLOGY

- DATA COLLECTION
 - DATA PREPROCESSING
 - SARIMA MODEL(A TIME SERIES ANALYSIS MODEL)
 - RANDOM FOREST(A MACHINE LEARNING MODEL)
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DATA COLLECTION



- We have gathered data on the total number of passengers traveling on domestic airlines in India from January 2012 to December 2022. The information was sourced from the Directorate General of Civil Aviation, and the dataset can be accessed through the following link: <https://www.dgca.gov.in/digigov-portal/?page=259/4184/servicename>
- To assess customer satisfaction, we acquired data from Kaggle, comprising over 20,000 observations. The dataset encompasses information from a randomly selected group of more than 20,000 individuals who have traveled on domestic flights in India. The dataset can be accessed via the following link: <https://www.kaggle.com/datasets/teejmahal20/airline-passenger-satisfaction?datasetId=522275>



WHY SARIMA MODEL USED?

- The utilization of time series modeling techniques, including ARIMA and SARIMA, emphasizes a focused exploration of temporal aspects. A detailed analysis of historical data provides insights into passenger flow, payload factors, and resource distribution. The reason for employing the SARIMA model lies in its ability to enhance predictive modeling through procedures such as data differencing, stationary checks, and a thorough autocorrelation analysis.
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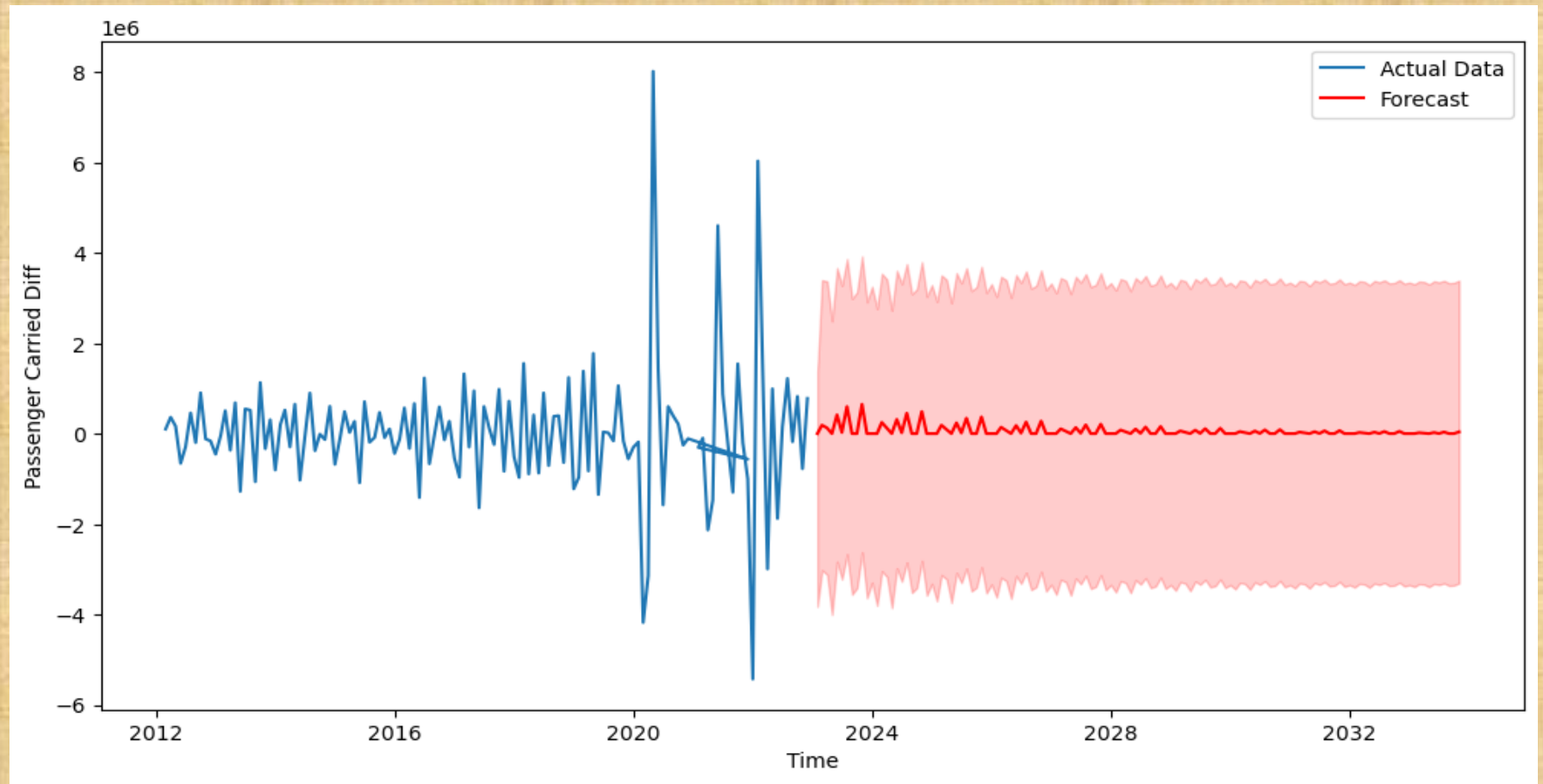
CODE FOR SARIMA MODEL

```
# Fit SARIMA model using statsmodels
sarima_model = SARIMAX(df['Passenger Carried Diff'], order=(1, 0, 1), seasonal_order=(1, 0, 1, 6))
sarima_fit = sarima_model.fit(dispatch=False)
# Make predictions
n_forecast = 130 # Number of periods to forecast
forecast = sarima_fit.get_forecast(steps=n_forecast)
# Set a lower bound for the predicted values
forecast.predicted_mean = np.maximum(forecast.predicted_mean, 0)
# Convert the index to DateTimeIndex
forecast_index = pd.date_range(start=df.index[-1], periods=n_forecast + 1, freq='M')[1:]
# Plot the results
plt.figure(figsize=(12, 6))
plt.plot(df.index, df['Passenger Carried Diff'], label='Actual Data')
plt.plot(forecast_index, forecast.predicted_mean.values, label='Forecast', color='red')
plt.fill_between(
    forecast_index,
    forecast.conf_int().iloc[:, 0],
    forecast.conf_int().iloc[:, 1],
    color='red',
    alpha=0.2
)

plt.xlabel('Time')
plt.ylabel('Passenger Carried Diff')
plt.legend()
plt.show()

# Calculate and print RMSE
actual_values = df['Passenger Carried Diff'][-n_forecast:].values
# Filter out zero values from the forecasted values
p_forecast = forecast.predicted_mean.values[forecast.predicted_mean.values > 0]
# Print the positive forecasted values separately
print(" Forecasted Values:")
print(p_forecast)
```

RESULT AND DISCUSSION



CODE FOR MACHINE LEARNING MODEL

```
as_df = pd.read_csv('airline_customer_satisfaction.csv')
#CHECKING FOR NULL VALUES
as_df.isnull().sum()
#SET THOSE NULL VALUES TO ZERO
as_df['Arrival Delay in Minutes'] = as_df['Arrival Delay in Minutes'].fillna(0)
#CHECKING FOR CLASS IMBALANCE
as_df['satisfaction'].value_counts()
as_df['satisfaction'].value_counts()/len(as_df)
#CHECKING FOR COLLINEARITY
from scipy.stats import chi2_contingency
chi2_contingency(pd.crosstab(as_df['Flight Distance'],as_df['Age']))
chi2_contingency(pd.crosstab(as_df['Gender'],as_df['Age']))
#CREATING TESTING AND TRAINING DATASET
finalpdframe = final_as_df.drop(labels=['Gender','Customer Type','Type of Travel','Class'], axis=1)
finalpdframe.head()
from sklearn.preprocessing import LabelEncoder
labelencoder = LabelEncoder()
finalpdframe['satisfaction'] = labelencoder.fit_transform(finalpdframe['satisfaction'])
X = finalpdframe.iloc[:, finalpdframe.columns != 'satisfaction']
y = finalpdframe.iloc[:, finalpdframe.columns == 'satisfaction']
X_train, X_test, y_train, y_test = train_test_split(X,y,test_size = 0.2, random_state = 1)
X.head()
#FITTING THE MODEL
from sklearn.ensemble import RandomForestClassifier
X_train.columns = X_train.columns.astype(str)#converting to sting format
rf = RandomForestClassifier(random_state=2, max_depth=3)
rf.fit(X_train, y_train.values.ravel())
# Save the model
import pickle
pickle.dump(rf,open("rf_model","wb"))
# Load the model
rf_loaded = pickle.load(open('rf_model', 'rb'))
print(confusion_matrix(y_test, y_pred))
print(accuracy_score(y_test, y_pred))
print(rf.score(X_test, y_test))
#VARIABLE IMPORTANCE
```



```

print(rf.score(X_test, y_test))
#VARIABLE IMPORTANCE
import pandas as pd
feature_importances = pd.DataFrame(rf.feature_importances_, index = X_train.columns,
                                   columns=['importance']).sort_values('importance', ascending=False)

feature_importances
#PLOTING VARIABLE IMPORTANCE
features = X.columns.values
importances = rf.feature_importances_
indices = np.argsort(importances)

plt.title('Feature Importances')
plt.barh(range(len(indices)), importances[indices], color='r', align='center')
plt.yticks(range(len(indices)), features[indices])
plt.xlabel('Relative Importance')
plt.show()
#RUNNING THE MODEL
def run_model(model, X_train, y_train, X_test, y_test, verbose=True):
    if verbose == False:
        model.fit(X_train, y_train, verbose=0)
    else:
        model.fit(X_train, y_train)
        y_pred = model.predict(X_test)
        roc_auc = roc_auc_score(y_test, y_pred)
        print("ROC_AUC = {}".format(roc_auc))
        print(classification_report(y_test, y_pred, digits=5))
        plot_confusion_matrix(model, X_test, y_test, cmap=plt.cm.Blues, normalize = 'all')

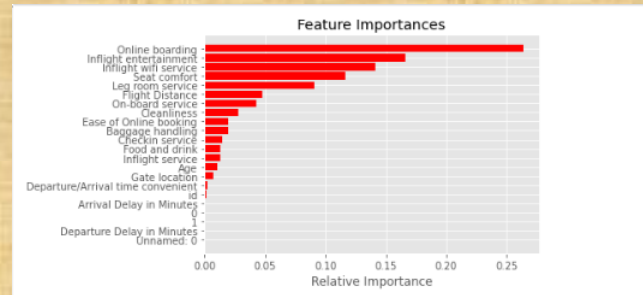
    return model, roc_auc
params_rf = {'max_depth': 25,
             'min_samples_leaf': 1,
             'min_samples_split': 2,
             'n_estimators': 100,
             'random_state': 42}

model_rf = RandomForestClassifier(**params_rf)
model_rf, roc_auc_rf = run_model(model_rf, X_train, y_train.values.ravel(), X_test, y_test.values.ravel())

```

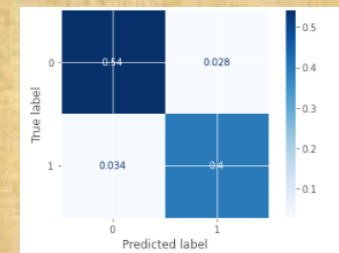
RESULTS

Variable Importance



Binary Classification Model Evaluation

ROC_AUC = 0.9361593459628734				
	precision	recall	f1-score	support
0	0.94155	0.95076	0.94613	2965
1	0.93370	0.92156	0.92759	2231
accuracy			0.93822	5196
macro avg	0.93762	0.93616	0.93686	5196
weighted avg	0.93818	0.93822	0.93817	5196





CONCLUSION

- In summary, the combination of SARIMA and random forest models highlights a significant upswing in domestic airline travel in India. Strategic enhancements targeting critical customer satisfaction factors, such as online boarding and in-flight amenities, are key to attracting a larger passenger base. The proposed plan, including route expansion and technology investments, sets the stage for long-term success. Both models predict substantial passenger growth in the next decade. Effective execution of these strategies positions airlines to capitalize on the increasing demand for domestic air travel in India.